

ECOM30001/90001 Basic Econometrics

Tutorial 3: Unbiasedness vs. Consistency of Estimators

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Semester 2, 2026

1. Unbiased Estimator (Finite Sample Property)

Definition: Unbiasedness

An estimator $\hat{\theta}$ of a population parameter θ is **unbiased** if its expected value equals the true parameter value for *any* sample size n :

$$E(\hat{\theta}) = \theta \quad \Longleftrightarrow \quad \text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta = 0$$

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Key Example: Sample Mean ($\bar{X}$)

- Estimator: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ where $E(X_i) = \mu_X$.
- Expected value:

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n}(n\mu_X) = \mu_X$$

- Since $E(\bar{X}) = \mu_X$, the sample mean $\bar{X}$ is an **unbiased estimator** of μ_X .

2. Consistency (Large Sample Property)

Definition: Consistency

An estimator $\hat{\theta}$ is **consistent** if $\hat{\theta} \xrightarrow{P} \theta$ as sample size $n \rightarrow \infty$.

Sufficient Condition (MSE Convergence):

$$\lim_{n \rightarrow \infty} \text{Bias}(\hat{\theta}) = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}) = 0$$

(Note: MSE convergence is a stronger condition than consistency; consistency does not guarantee finite-sample unbiasedness.)

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Intuition

- An estimator can be **biased in finite samples**, but still **consistent** if the bias disappears as $n \rightarrow \infty$!

(b) Standard Sample Mean ($\bar{X}$)

Setup

Consider the standard sample mean estimator of population mean μ_X :

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Statistical Properties:

- **Unbiasedness:** $E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \mu_X \implies \text{Bias}(\bar{X}) = 0$
- **Variance & Asymptotics:** $\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \xrightarrow{n \rightarrow \infty} 0$

Conclusion

Unbiased and Consistent since $E(\bar{X}) = \mu_X$ for any n , and $\text{Var}(\bar{X}) \rightarrow 0$ as $n \rightarrow \infty$.

(c) Shifted Estimator (X^*)

Setup

Consider adding 1 to the numerator of the sample sum:

$$X^* = \frac{\sum_{i=1}^n X_i + 1}{n} = \bar{X} + \frac{1}{n}$$

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Statistical Properties:

- **Expectation & Bias:**

$$E(X^*) = E\left(\bar{X} + \frac{1}{n}\right) = \mu_X + \frac{1}{n} \implies \text{Bias}(X^*) = \frac{1}{n} \neq 0$$

- **Variance & Limits:**

$$\text{Var}(X^*) = \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \xrightarrow{n \rightarrow \infty} 0, \quad \lim_{n \rightarrow \infty} \text{Bias}(X^*) = 0$$

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Conclusion

Biased in finite samples, but Consistent since both $\text{Bias} \rightarrow 0$ and $\text{Var} \rightarrow 0$ as $n \rightarrow \infty$.

(g) OLS Intercept Bias

Setup

True DGP: $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ ($\beta_0 = 0.5, \beta_1 = 1, E(X_i) = \mu_X = 0.25$).

Misspecified Intercept-Only Model: $Y_i = \gamma_0 + \epsilon_{3i} \implies$ OLS estimator is $g_0 = \bar{Y}$. (Recall last tute)

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Mathematical Derivation of Bias $B = E(g_0) - \beta_0$:

$$\begin{aligned} B &= E(g_0) - \beta_0 = E(\bar{Y}) - \beta_0 \\ &= E\left(\frac{1}{n} \sum_{i=1}^n (\beta_0 + \beta_1 X_i + \epsilon_{1i})\right) - \beta_0 \\ &= E(\beta_0 + \beta_1 \bar{X} + \bar{\epsilon}_1) - \beta_0 \\ &= \beta_0 + \beta_1 E(\bar{X}) + E(\bar{\epsilon}_1) - \beta_0 \quad (\text{since } E(\bar{\epsilon}_1) = 0) \\ &= \beta_1 E(\bar{X}) = (1)(0.25) = \mathbf{0.25} \quad (\text{since } \beta_1 = 1, E(\bar{X}) = 0.25) \end{aligned}$$

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Conclusion

Omitting X_i biases the intercept by $B = \mathbf{0.25}$, matching the simulation!

(MCQ) Scaled Estimator ($\tilde{X}$)

Setup: biased? Consistent?

Consider adding n to each observation before averaging:

$$\tilde{X} = \frac{\sum_{i=1}^n (X_i + n)}{n}$$

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 $\text{Var}(\tilde{X}) = \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \xrightarrow{n \rightarrow \infty} 0, \quad \lim_{n \rightarrow \infty} \text{Bias}(\tilde{X}) = \infty$

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Conclusion

Biased and Inconsistent since the bias explodes to ∞ as $n \rightarrow \infty$.